



Fonds National de la
Recherche Luxembourg



MINIMAL MODEL FOR THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS

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671. Heraeus Seminar

- **Nonequilibrium phase transitions** are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- **Stochastic thermodynamics** provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ **thermodynamics of nonequilibrium phase transitions**.
- **Practical motivation** to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
 - Limitations of energy-converting nanomachines $\Rightarrow$ Consider ensembles of nanomachines
 - Phase transitions affect performance of large ensembles of machines?
 - Role of the number of nanomachines?
- **Synchronization** as a promising candidate for a beneficial performance of the ensemble.

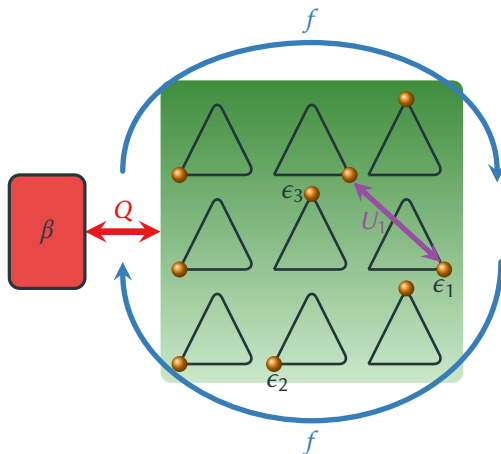
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 - Phase transitions affect performance of large ensembles of machines?
 - Role of the number of nanomachines?
- **Synchronization** as a promising candidate for a beneficial performance of the ensemble.

$\Rightarrow$ Search for a (feasible) thermodynamic model exhibiting synchronization.

[Herpich, Thingna, & Esposito arXiv:1709.06045]

OPEN THREE-STATE UNIT NETWORK

- **Network** of N all-to-all coupled three-state units [PRL **96**, 145701 (2006)] units.
- Open system in contact with **heat bath**.
- **Interaction energy** $U(\mathbf{N}) = \frac{u}{2N} \sum_{k=1}^3 N_k^2 + U_0$.
- Subjected to **torque** f acting as a global bias.



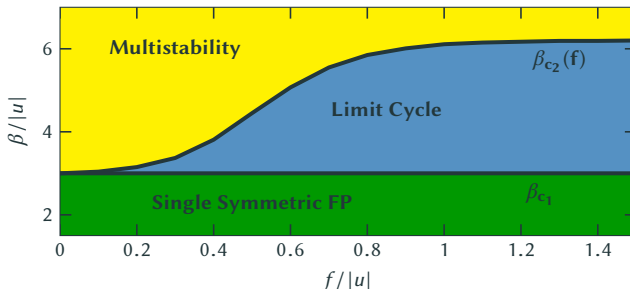
- **Markovian master equation** with Arrhenius rates satisfying **local detailed balance**

$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N}, \mathbf{N}'} P_{\mathbf{N}'}, \quad \ln \frac{W_{\mathbf{N} \mathbf{N}'}}{W_{\mathbf{N}' \mathbf{N}}} = -\beta [\Delta F^{eq}(\mathbf{N}, \mathbf{N}') - \sigma(\mathbf{N}, \mathbf{N}') f], \quad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1} \overbrace{\ln \frac{N!}{\prod_i N_i!}}^{S^{int}(\mathbf{N})}$$

- Mean-field dynamics given by a nonlinear master equation

$$\dot{\bar{n}}_i(t) \equiv \langle \dot{n}_i(t) \rangle = \sum_j k_{ij}(\bar{n}_i, \bar{n}_j) \bar{n}_j(t), \quad \ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \sigma(i, j) f].$$

- For $\epsilon_i = \epsilon$, the nonlinear equation has a **symmetric fixed point** $\dot{\bar{n}}_i = \sum_j k_{ij} \bar{n}_j \big|_{\bar{n}_j=1/3} = 0$.
- Linear stability indicates **Hopf bifurcation**: $\lambda_{\pm} = -\Gamma(\beta u + 3) \cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\Gamma \sinh\left(\frac{\beta f}{2}\right)$.



- One equilibrium (first-order) and two nonequilibrium phase transitions at $\beta = \beta_{c1,2}$.
- Three distinct fixed points in the **low-temperature regime** ($\beta \gg \beta_c$) **Energy-driven**
- Symmetric solution in the **high-temperature regime** ($\beta \ll \beta_c$) **Entropy-driven**
- Thermodynamic consistency constraints **stable oscillations** to emerge only at intermediate temperatures ($\beta_{c1} < \beta \leq \beta_{c2}$).

⇒ Minimal thermodynamic model exhibiting synchronization

Stochastic Dynamics

- The microscopic/mesoscopic dynamics is ruled by an irreducible and Markovian master equation $\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'} P_{\mathbf{N}'}$ and exhibits a unique steady state (PF theorem).

Mean-Field Dynamics

- The macroscopic dynamics is governed by a nonlinear master equation $\dot{\bar{n}}_i = \sum_j k_{ij} \bar{n}_j$ and displays stable oscillations ($\beta_{c1} \leq \beta < \beta_{c2}$) and multistability ($\beta \geq \beta_{c2}$).

⇒ **Non-commutation of the thermodynamic- and the stationary limit.**

$$\lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \mathbf{P}(t) \neq \lim_{N \rightarrow \infty} \underbrace{\lim_{t \rightarrow \infty} \mathbf{P}(t)}_{\mathbf{P}^s}, \quad \text{if } \beta \geq \beta_{c1}.$$

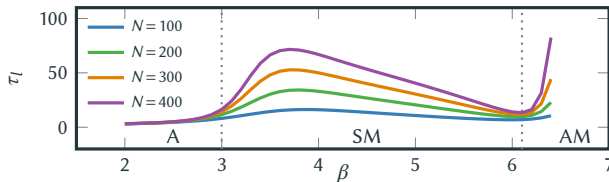
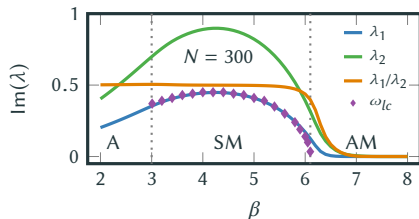
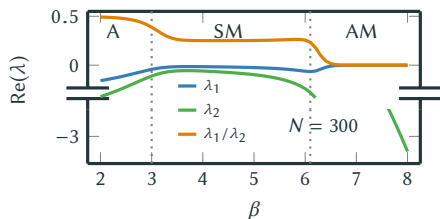
- Apparent paradox since mean field represents the exact solution in the $N \rightarrow \infty$ limit.
- Idea:** Infinite-time solutions of mean-field exhibited also by finite systems at long but *finite* timescales → **Metastability?**

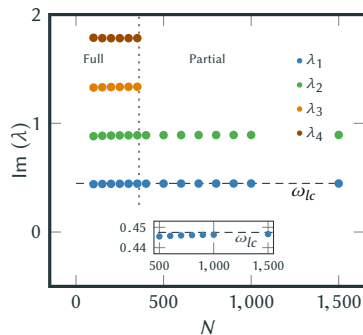
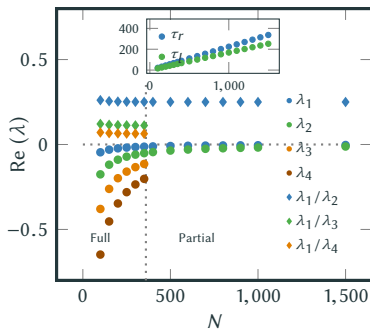
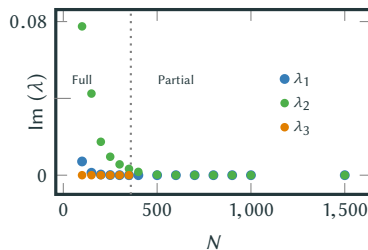
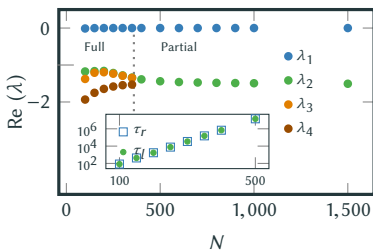
- Formal solution of the finite-size master equation reads

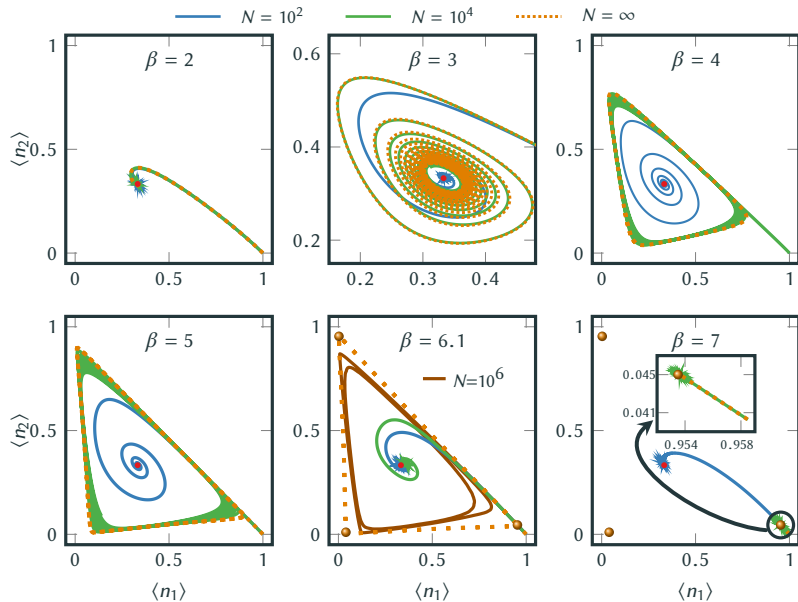
$$\mathbf{P}(t) = e^{\mathbf{W}t} \mathbf{P}(0) = \sum_{i, i^*} \underbrace{\Phi_i^L \mathbf{P}(0)}_{\equiv c_i} e^{\lambda_i t} \Phi_{i^*}^R + c_{i^*} e^{\lambda_{i^*} t} \Phi_{i^*}^R,$$

- Relevant timescales are specified by the slowest decaying modes.

$$\text{Relaxation } \tau_r \sim -\frac{1}{\text{Re}(\lambda_1)}, \quad \text{Metastable } \tau_m \sim -\frac{1}{\text{Re}(\lambda_2)}, \quad \text{Lifetime } \tau_l \equiv \tau_r - \tau_m \sim \frac{\text{Re}(\lambda_1) - \text{Re}(\lambda_2)}{\text{Re}(\lambda_1) \text{Re}(\lambda_2)},$$



$\beta = 4.0$ (SM phase) $\beta = 7.0$ (AM phase)

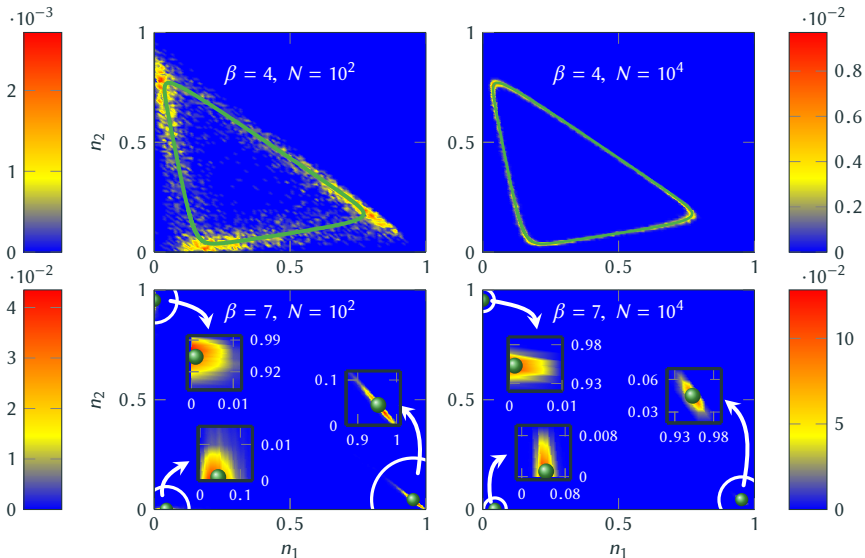


$\Rightarrow$ Large system dynamics indeed agrees with the mean-field solution at metastable times.

ROLE OF INITIAL CONDITION FOR METASTABILITY I

- Recall that $\mathbf{P}(t) \stackrel{t \gg \tau_m}{\approx} c_0 \Phi_0^R + e^{\lambda_1 t} c_1 \Phi_1^R + e^{\lambda_1^* t} c_1^* \Phi_1^{R*}$ with $c_{1,1^*} = \Phi_{1,1^*}^L \cdot P(0)$.
- Metastability can be removed from the dynamics for $P(0) = b_0 \Phi_0^{R*} + \sum_{i \neq 1, 1^*} b_i \Phi_i^{R*}$.

Is metastability a generic property of our system or merely an artifact of special initial conditions?



$$N = 10^2$$

$$\beta = 4$$

$$N = 10^4$$

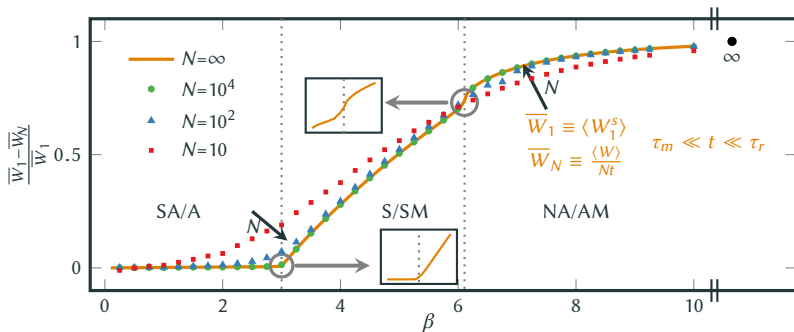
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DISSIPATED WORK

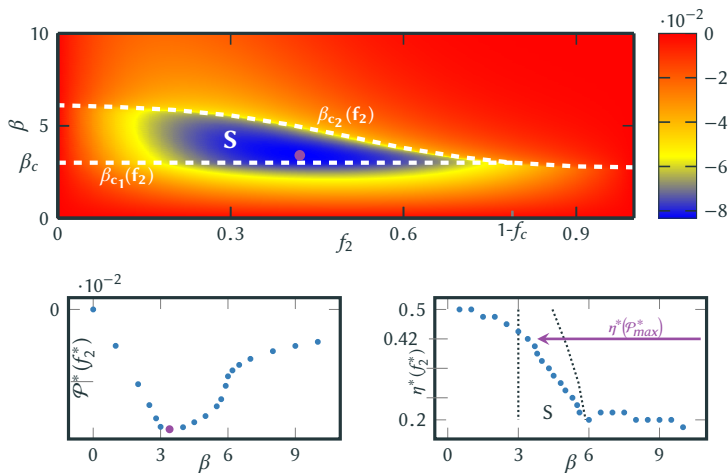
- **First Law:** $d_t \langle E \rangle = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$
- With this setup, work needs to be done on average to maintain the system, $\langle \dot{W} \rangle > 0$, which is on average dissipated as heat into the bath, $\langle \dot{Q} \rangle < 0$.



- **(Signatures of) Nonequilibrium phase transitions** at the critical points (for large but finite N). [“First-order phase transition” at β_{c1} and “second-order phase transition” at β_{c2}].
- **Reduction** of the dissipated work for **finite interactions** $u < 0$.
- Dissipated work is **further reduced** for **smaller** networks in the two high-temperature regimes.

EFFICIENCY AT MAXIMUM POWER

- **Work-to-work conversion** $f \equiv f_1 - f_2$, with input work $\mathcal{W}_1(f_1)$ and output work $\mathcal{W}_2(f_2)$.
- **Maximization of the output power** $\mathcal{P} \equiv \dot{\mathcal{W}}_2$ with respect to output force $\partial_{f_2} \mathcal{P}|_{f_1=1} = 0 \Rightarrow f_2^*$.
- Efficiency of work-to-work conversion at maximum power $\eta^* \equiv 1 - \frac{f}{f_1}|_{f_2=f_2^*}$



- Power output is maximized in the **synchronization** phase.
- The **efficiency at maximum power**, $\eta^*(f_2^*)$, is surprisingly **close** to the universal value, $\eta^* = 1/2$, for a tightly-coupled (single net-current) system responding linearly.

- Minimal thermodynamic model exhibiting synchronization.
- Model with two non-equilibrium and one equilibrium phase transition.
- Resolving the apparent paradox of a finite irreducible Markov process and mean-field dynamics displaying stable oscillations and multistability via notions of metastability.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- When operating as work-to-work converter, the maximum output is achieved for a synchronous operating mode. The associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

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Thank you for your attention!